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LATE CRETACEOUS-EARLY TERTIARY PALEOMAGNETISM OF ARUBA AND BONAIRE (NETHERLANDS LEEWARD ANTILLES)

Carola Stearns, Frederick J. Mauk, and Rob Van der Voo

Department of Geological Sciences, University of Michigan, Ann Arbor, Michigan 48109

Abstract. For a paleomagnetic study of the Upper Cretaceous of the Netherlands Leeward Antilles, we have analyzed 187 oriented samples from the complex norite-tonalite batholith and the diabase-schist-tuff formation on Aruba and 44 oriented samples from various igneous units of the Washikemba Formation on Bonaire. Both alternating field and thermal demagnetization procedures were used to demagnetize the samples. Most of the cores yielded univectorial Zijderveld diagrams above 10-20 mT or 400° C. Virtual geomagnetic poles calculated from site means were then compared with equivalently aged poles for cratonic South America and other Caribbean sites. The Washikemba Formation of Bonaire and the diabase-schist-tuff formation of Aruba yield poles that are similar to those of Guajira (Colombia) and the Caribbean Mountains of northern Venezuela, whereas they are rotated approximately 90° with respect to the pole for stable South America. The batholith on Aruba yields a pole which is rotated clockwise over approximately 20° with respect to the South American pole. Thus, our data strongly support the hypothesis that Aruba and Bonaire have been rotated clockwise by as much as 90° since the Late Cretaceous. In addition, paleolatitudes calculated from the inclinations of the paleomagnetic vectors suggest that there was a northward drift with respect to South America of perhaps as much as 10° of the islands during the rotation, although this can hardly be called a statistically significant amount.

Introduction

To explain a variety of geological and geophysical data, Silver et al. [1975] proposed that Aruba and Bonaire were part of a microplate, the Bonaire Block, which has had complex interactions with the South American and Caribbean macroplates (see Figure 1). If the relative motions of North and South America proposed by Ladd [1976] are correct, then the evolution of the southern Caribbean involved convergent and divergent as well as transcurrent motions. Both Aruba and Bonaire are characterized by radiometrically and paleontologically dated episodes of igneous activity from the Albian through the Paleocene and thus are well suited for a paleomagnetic investigation of possible latitudinal and/or rotational tectonics associated with the Bonaire Block.

Previously published paleomagnetic work on similarly aged rocks from other parts of the Bonaire Block include data from the Guajira Peninsula of Colombia [MacDonald and Opdyke, 1972] and the Caribbean Mountains of northern Venezuela [Skerlec and Hargraves, 1980]. Both studies yield paleomagnetic directions with equatorial or subequatorial pole positions for the microplate and rotated approximately 90° away from calculated mean poles for contemporaneous cratonic regions of both North and South America [Van der Voo and French, 1974; Creer, 1962, 1970; Valencio, 1972; Opdyke and MacDonald, 1973; Pacca and Hiodo, 1976; Vilas, 1976; Valencio et al., 1977; Mendia, 1978]. Although some earlier workers [Watkins and Cambray, 1970; Fink and Harrison, 1971; Guja and Vincenz, 1971; Vincenz et al., 1973] have interpreted similar 'anomalous' data as reflecting nondipolar magnetic field complexities related to polarity reversals, others [MacDonald and Opdyke, 1972; Dasgupta and Vincenz, 1975; Gose and Schwartz, 1977; Skerlec and Hargraves, 1980] prefer to explain the data as the result of tectonic rotation. Skerlec and Hargraves suggest that the increasing number of consistent paleomagnetic results derived from widely separated sites within the Caribbean realm combined with the probable short duration of a magnetic reversal [McElhinny, 1973] indicates that tectonic rotation is a more likely explanation of the data.

Our study involved 231 samples from 24 sites in Late Cretaceous and early Tertiary rocks from Aruba and Bonaire. We interpret the data as additional support for an approximately 90° rotation of the Bonaire Block or at least segments of that block. Furthermore, we suggest that our data are evidence for a clockwise sense to the rotation.

Regional Geologic Setting

The Leeward Antilles form an east-west trending chain of small islands and atolls located off the northern coast of Venezuela in the southern part of the Caribbean. The island chain consists of two segments displaced in a north-south sense by approximately 100 km at 66° W longitude. The northwesternmost segment of the chain begins at the island of Aruba and continues eastward to La Orchilla. The southeasternmost segment begins at the island of Tortuga and continues eastward to Tobago (see Figure 2). Our study concerns the islands of Aruba and Bonaire, part of the westernmost segment of the Leeward Antilles Group, but does not include Curacao, where we did not sample.

Structurally, the offshore region is characterized by east-west trending folds with superimposed northwest-southeast trending normal faults [Silver et al., 1975]. Detailed descriptions of these structures based on gravity, magnetic, and seismic profiles have been given by Lagaay [1969] and Silver et al. [1975]. Aruba, Bonaire, and Curacao sit at the crest of one of these block-faulted anticlines and are paralleled to the north by the Los Roques Trough and the Curacao Ridge and to the south by the Bonaire Basin.

Geology of the Islands

The geology of the islands has been discussed by many authors including Westerman [1932], Lagaay [1969], Beets [1972], Maresch [1974], Silver et al. [1975] and Beets and MacGillavry [1977]. Simplified

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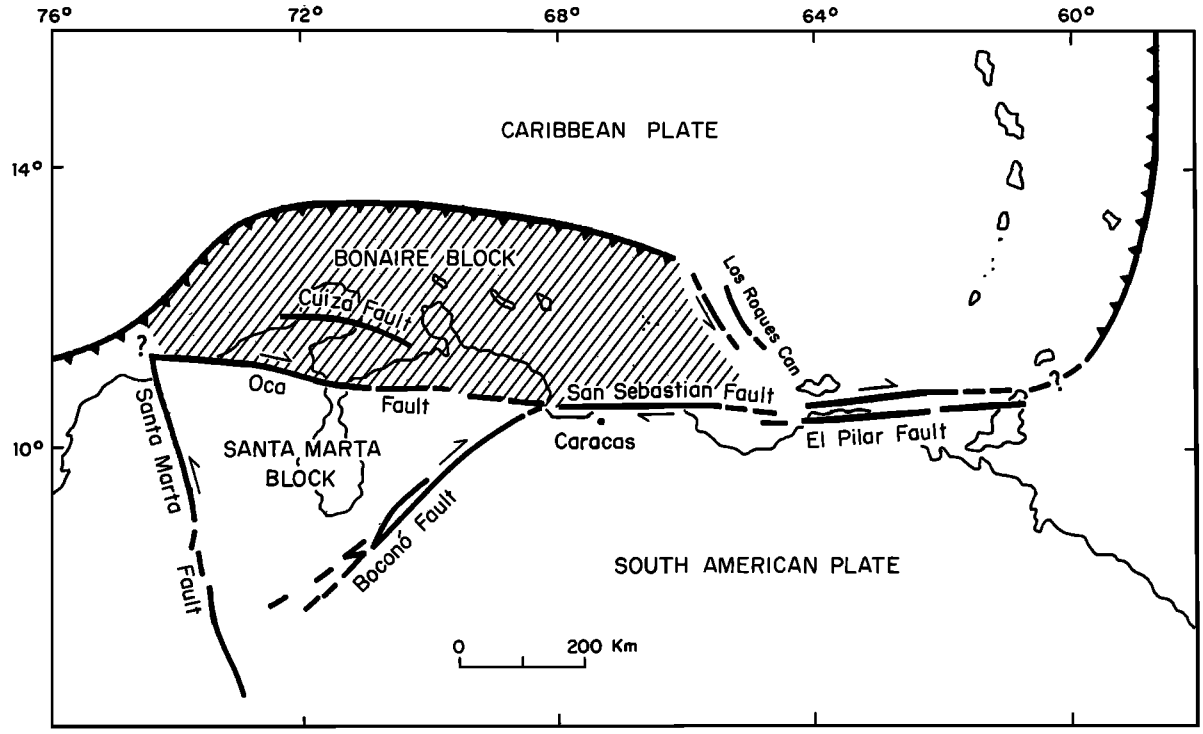


Fig. 1. Map of the Bonaire Block (shaded) [after Silver et al., 1975].

geologic maps of northwestern Bonaire and of Aruba are given in Figures 3 and 4, respectively. Beets and MacGillavry [1977] divided the geologic history of the islands into two periods. The first, during the Cretaceous and early Tertiary, is characterized by strong magmatic activity and mobility. In contrast, the mid-Eocene to Holocene history is one of relative stability as seen from the lack of any younger than mid-Eocene igneous rocks on the islands [Silver et al., 1975].

The Cretaceous and early Tertiary history can be further subdivided into two phases. The first phase, from the Aptian through the Coniacian ($\sim$ 110-85 m.y.), was marked by abundant submarine volcanism [Beets and MacGillavry, 1977]. On Bonaire this is represented by the Washikemba Formation, which includes numerous intermediate to mafic volcanic rocks and shallow intrusives, and on Aruba by the diabase-schist-tuff formation, sometimes called the Older Formation, the igneous units of which are primarily basalts. The second phase, from the Campanian through the early Paleocene ($\sim$ 75-60 m.y.) shows a decrease in magmatic activity and an increase in clastic sedimentation on Bonaire [Beets and MacGillavry, 1977]. However, no sedimentary strata of this age have been found on Aruba, and it was during this time that the complex batholith was emplaced.

The Cretaceous and early Tertiary igneous nuclei of Aruba and Bonaire are overlain by Eocene to Holocene limestones and alluvial sediments [Herweijer et al., 1977]. The islands are rimmed by extensive Quaternary limestones which have been studied by De Buisonjé [1964, 1974]. De Buisonjé suggests that the history of these formations is best explained by a steady tectonic uplift superimposed on Quaternary sea level fluctuations. The altitudes of the highest, oldest limestone terraces and erosional surfaces indicate that the amount of tectonic uplift is of the order of several hundred meters [Silver et al., 1975].

The ages of the Washikemba Formation of Bonaire and the diabase-schist-tuff formation of Aruba are based primarily on their biostratigraphy. Beets and MacGillavry [1977] report that fossil ammonites found near the base of the Washikemba Formation are mid-Albian and that inoceramic and globotruncanid faunas near the top of the formation are Coniacian. The diabase-schist-tuff formation is thought to be roughly the time equivalent of the Washikemba Formation. This is supported by some possible Turonian ammonites found by W. D. MacDonald within a sedimentary unit (quoted by Beets and MacGillavry [1977]).

In addition to the biostratigraphic ages, radiometric dates have been published for the Washikemba Formation [Priem et al., 1979] and for the tonalite batholith on Aruba [Priem et al., 1978]. Because the K-Ar dates determined for the Washikemba Formation, 78±2 (Campanian) and 61±4 (Paleocene), are significantly younger than the biostratigraphic age of the formation, Priem et al. [1979] interpret them as reflecting two separate low-grade metamorphic events. Priem et al.'s [1978] Rb-Sr dates for the batholith on Aruba place the main intrusion at 70.4±2.0 m.y. However, they also suggest that there is evidence for an earlier intrusive phase at 85.1±.5 m.y. Field reconnaissance strongly supports the idea of a complex, multistage intrusion.

In summary, the basement rocks of the Netherlands Antilles are primarily Early Cretaceous pillow basalts and diabases which are overlain by Late Cretaceous marine sedimentary rocks and intruded by dioritic stocks [Beets and MacGillavry, 1977; Silver et al., 1975]. These, in turn, are overlain by Eocene to Holocene limestones and alluvial sediments [Herweijer et al., 1977]. Although

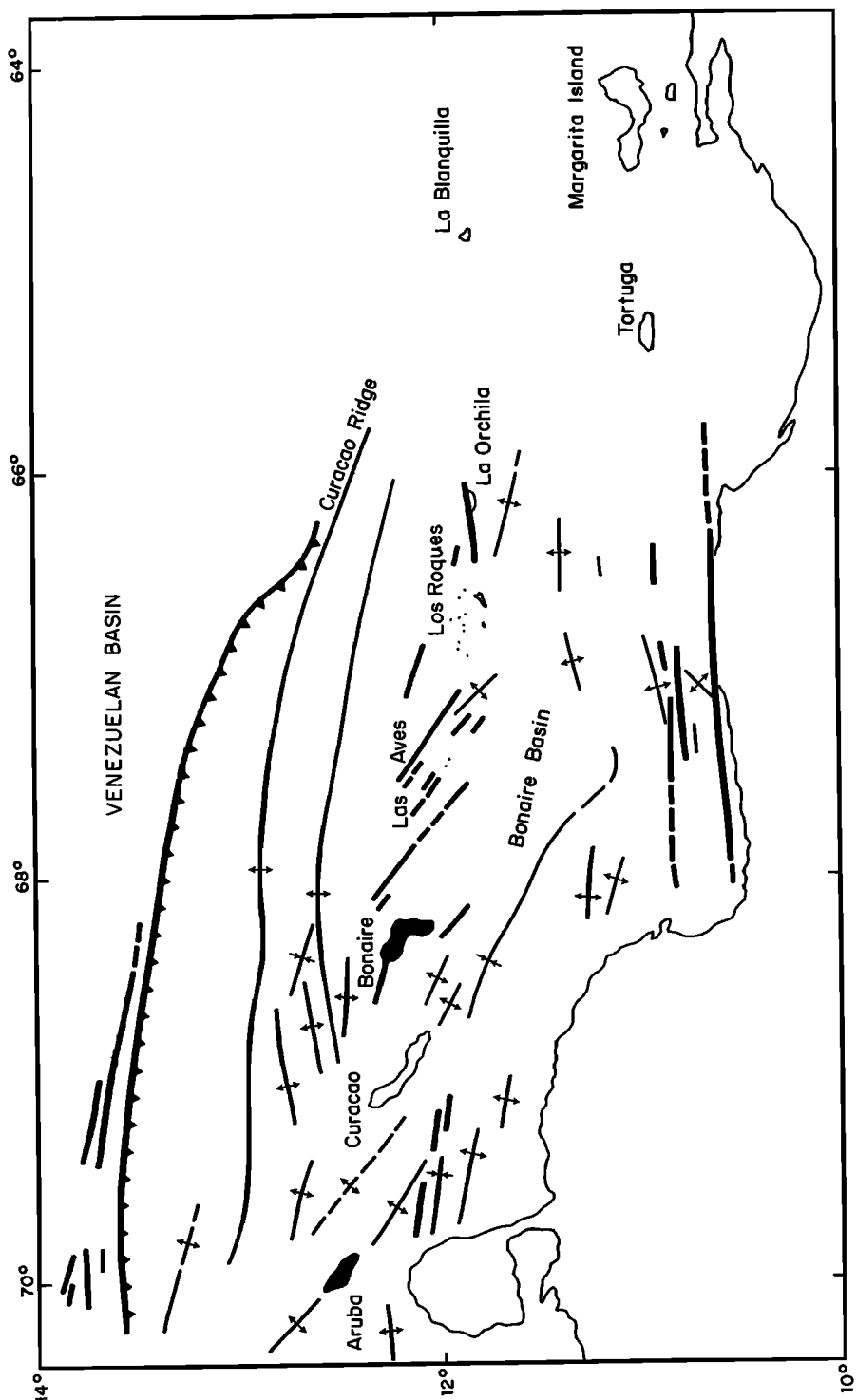


Fig. 2. Regional geology of the Venezuelan borderland [after Silver et al., 1975]. Aruba and Bonaire are highlighted in black.

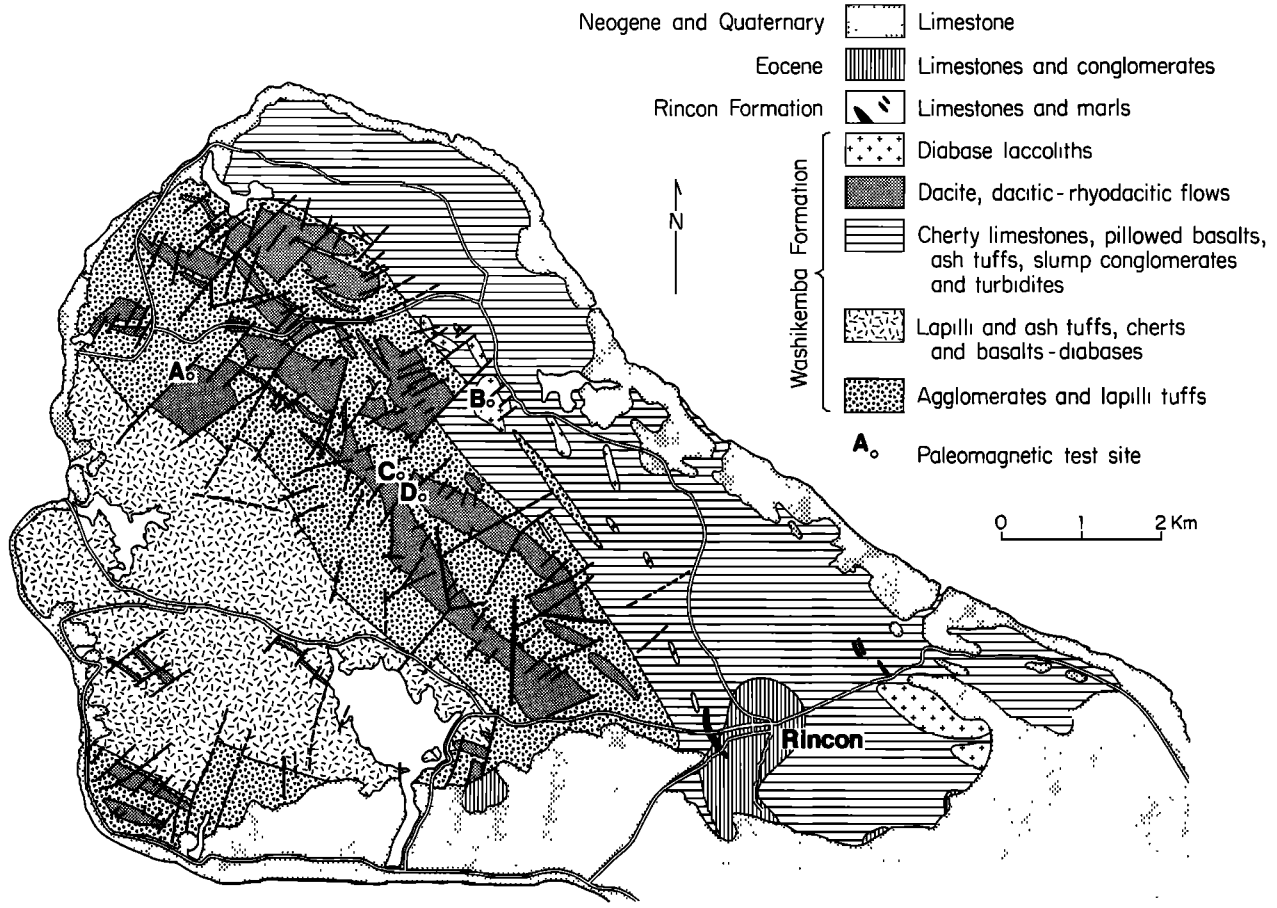


Fig. 3. Simplified geological map of northwestern Bonaire, showing sampling localities (letters), after unpublished data from G. Klaver [Beets et al., 1977].

relatively contemporaneous rocks can be identified on all of the Netherlands Antilles, the rocks are not directly correlative.

Sampling and Laboratory Treatment

Samples were collected from four sites representing various igneous units of the Washikemba Formation on Bonaire. All of the samples were taken from within the National Park on the northwest corner of the island. Locations are illustrated in Figure 3. The Washikemba Formation is thought to be a typical island arc succession consisting primarily of basalts, andesites, pyroclastics, pelagic cherts, cherty limestones, conglomerates, and sandstones [Beets et al., 1977]. Four of the igneous units were sampled: a diabase near Salina Matijs (locality B, Figure 3), a dacite from the slopes of Brandaris (locality A, Figure 3), a rhyodacite flow (locality D), and a tuff (locality C).

Oriented samples were collected from the 11 localities on Aruba which are shown in Figure 4. Localities A-H are located in the complex batholith. Localities K-M are located within the diabase-schist-tuff formation. This formation is a low-grade metamorphosed sequence of predominantly basalts and diabases [Helmers and Beets, 1977]. Three units were sampled near Dos Playa and Boca Prins: a pillow basalt (locality M of Figure 4), an andesite porphyry (locality L), and a conglomerate made of well-rounded basalt clasts in a hematite-stained matrix (locality K). Unfortunately, the stratigraphy of this formation is poorly known, and the precise statigraphic relationships of these units are not known. Sampling of the batholith was carried out along a 7-km stretch on the northeast coast of Aruba. Although the batholith has been called a norite-tonalite, the lithology actually varies from a quartz-norite to a quartz-hornblende-gabbro [Helmers and Beets, 1977]. Numerous felsic and mafic dikes crosscut the intrusion. In addition to the batholith itself, we collected samples from several crosscutting felsic dikes (localities D, F, and G), which are possibly comagmatic [Helmers and Beets, 1977]. Generally, structural corrections were not possible either because of poor outcrop exposure or because the samples were taken from the batholith. We were able to measure the attitude (strike ∿ 60°W, dipping 90°) of the conglomerate (site 19, locality K) in the diabase-schist-tuff formation on Aruba, however.

Beets et al. [1977] describe the exposure of the Washikemba Formation in northwestern Bonaire as a 'gently dipping section.' If true, any structural corrections might not significantly alter the data. Our own observations in general confirmed the low dip angles, although steeper dips could occasionally be surmised. The issues related to the structural corrections will be discussed further below.

Natural remanent magnetization (NRM) characteristics were measured for all the samples using an ScT cryogenic magnetometer or a Schonstedt SSM-1A spinner magnetometer. To isolate secondary and characteristic components of the NRM, stepwise

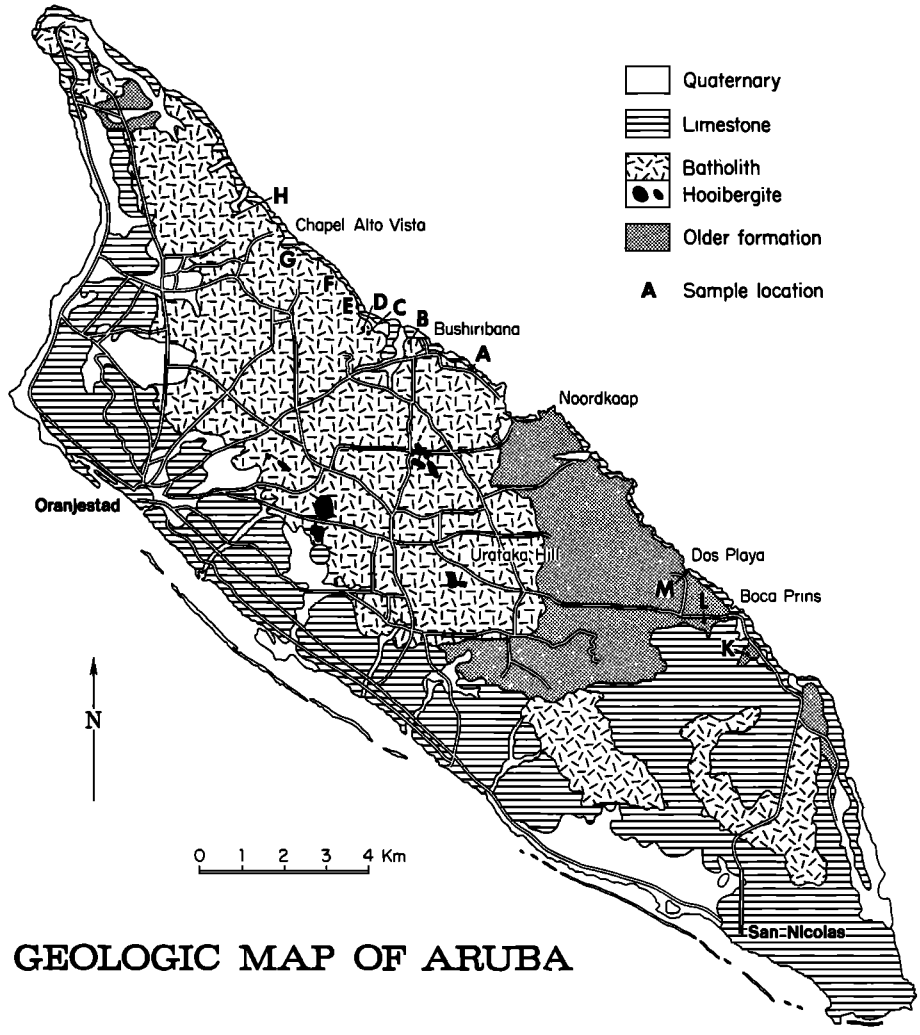


Fig. 4. Simplified geological map of Aruba showing sampling localities (letters) [after Westerman, 1932].

demagnetization procedures, both alternating field (af) and thermal, were done on at least six samples per site. The demagnetization results were then plotted on orthogonal projections such as those shown in Figures 5-8 [Zijderveld, 1967]. Treatment was carried out until it was evident that all components within the samples had been identified. Directions of magnetization were determined by vector subtraction of straight-line demagnetization trajectories generally defined by three or more points on the Zijderveld diagrams. Only those results that we interpreted as characteristic were used in calculating mean directions. Virtual geomagnetic poles (VGP) were calculated from site means in order to compare our data with data from other Caribbean sites.

Results

Washikemba Formation, Bonaire

Typical Zijderveld demagnetization diagrams for three of the lithologies sampled in the Washikemba Formation are illustrated in Figure 5. In general, all of the samples exhibited one or two components of magnetization defined by at least three colinear points on the Zijderveld plots. Above 15 mT or 400°C, the demagnetization trajectories were univectorial. Site means are plotted on an equal area projection in Figure 9a and listed with their statistics in Table 1.

Two demagnetization diagrams for the diabase (locality B of Figure 3) are illustrated in Figures 5a and 5b. All of the samples revealed two approximately antiparallel directions of magnetization separable by both alternating field and thermal procedures. Because the direction first removed has a low coercivity ($\leq$15 mT) as well as low blocking temperatures ($\leq$400°C), it is considered to be either an antiparallel remanent magnetism carried by a magnetic mineral other than magnetite or a secondary direction acquired during a reversed magnetic period. The second component, with a coercivity $\leq$100 mT and blocking temperatures between 500° and 585°C, had better resolution than the first removed component in both demagnetization techniques ($k = 90$ versus $k = 29$) and so was used to calculate the site mean (D/I = 104/-21).

The magnetization of the rhyodacite (Figure 5c; locality D of Figure 3) did not decay in alternating field procedures up to 100 mT. Thermal demagnetization of the specimens produced univectorial Zijderveld diagrams like the one shown in Figure 5c.

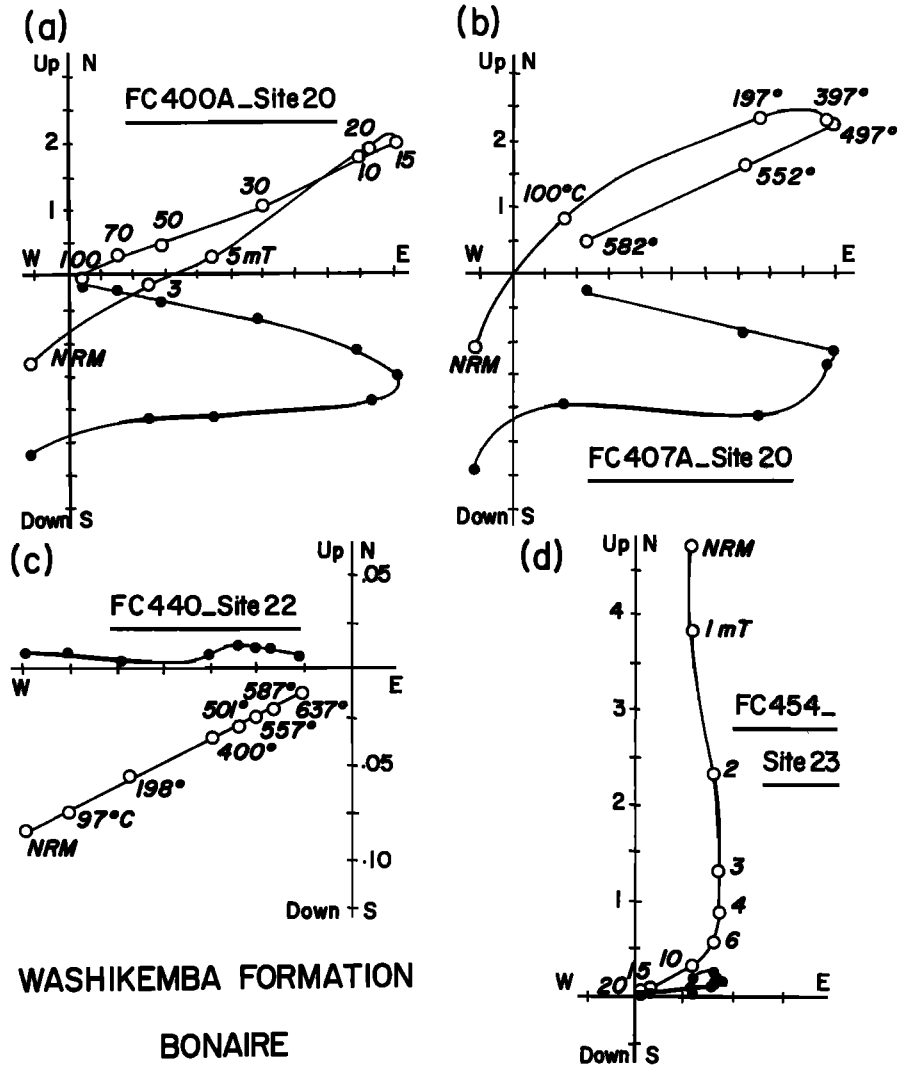


Fig. 5. Demagnetization diagrams [Zijderveld, 1967] of samples from the Washikemba Formation of Bonaire. Solid circles represent projections on the horizontal plane, and open circles those on the vertical plane. Units indicated for the axes are in A m^{-1} (1 A m^{-1} = 10^{-3} emu/cm^3). (a) An af demagnetization of a diabase from locality B. (b) Thermal demagnetization of the same. (c) Thermal demagnetization of the rhyodacite from locality D. (d) An af demagnetization of dacite from locality A.

The maximum blocking temperatures of this magnetization were 650° C, indicative of hematite. Because the site mean (D/I = 273/+21) is nearly antiparallel to those of sites A and B, the magnetization was probably acquired during a period of opposite polarity to that recorded at sites A and B. (Note that Aruba and Bonaire were located near the paleoequator for the Cretaceous, and because of the large number of magnetic reversals in the Late Cretaceous and early Tertiary [Lowrie and Alvarez, 1977; Roggenthen and Napoleone, 1977] as well as the lack of precision in the ages of our samples, it is virtually impossible to determine which of the magnetizations were acquired during a 'normal' or 'reversed' period per se. In Figure 9 we plotted all of the site means in the eastern hemisphere for convenience. The mean directions actually measured are listed in Table 1.)

A typical Zijderveld diagram for the dacite from Brandaris (locality A of Figure 3) is shown in Figure 5d. All of the samples showed a prominent first removed direction. We interpret this nearly vertically upward component as a viscous magnetization possibly acquired during sample storage. Above 10 mT or 400° C, the samples yielded univectorial Zijderveld diagrams. This stable magnetization clustered very well (k = 386) and is similar to the mean direction of the diabase.

The tuff (locality C of figure 3) showed less consistent behavior. Although most of the samples yielded univectorial Zijderveld diagrams, the NRM of 28 samples did not cluster very well (k = 4.6). After demagnetization, the statistics for the 11 analyzed samples did not improve (k = 4.3), and we do not feel that these data are reliable. The mean direction for the 11 samples (D/I = 26/-30) is not similar to the mean directions yielded by the other lithologies.

As the equal-area projection of the site means in Figure 9a illustrates, three out of the four lithologies from the Washikemba Formation yield characteristic magnetic directions with easterly decli-

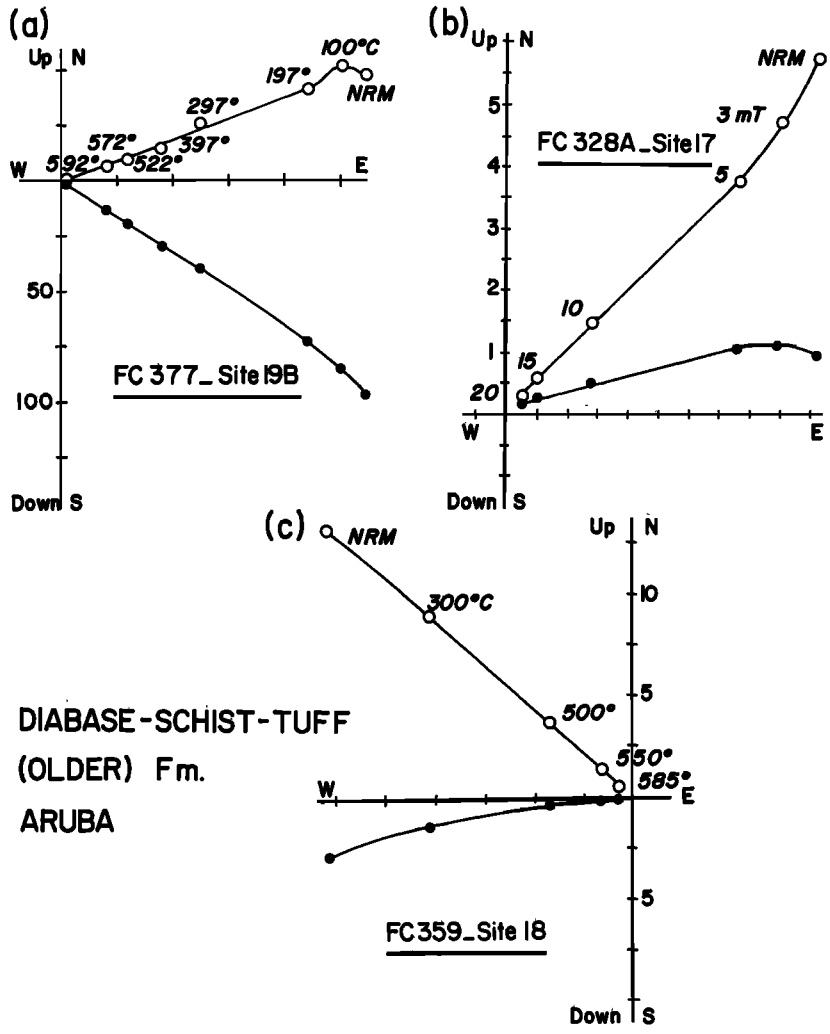


Fig. 6. Demagnetization diagrams (as in figure 5) for samples from the Older Formation of Aruba (diabase-schist-tuff formation). (a) Thermal demagnetization of the conglomerate from locality K. (b) An af demagnetization of the pillow basalt from locality M. (c) Thermal demagnetization of the andesite porphyry from locality L.

nations and shallow negative (up) inclinations. These sites showed good internal behavior, as revealed by the large precision parameters and small α_{95} values listed in Table 1. The data from the anomalous site (locality C of Figure 3) were the least consistent and will be disregarded.

Diabase-Schist-Tuff Formation (Older Formation), Aruba

Zijderveld diagrams for each of the sampled lithologies in the diabase-schist-tuff formation are shown in Figure 6. Most samples from sites 18, 19A, and 19B (localities L and K of Figure 4) exhibited stable and often univectorial magnetizations, as is illustrated by the straight demagnetization trajectories in Figures 6c and 6a, respectively. The behavior of the samples from site 17 was more complex. Some of the samples yielded near-univectorial plots (Figure 6b), and some revealed secondary components. For these reasons we have selected those samples that yield 'characteristic' magnetic directions on the basis of three or more colinear points for the calculation of the mean direction for these sites. Fifteen of the twenty demagnetized samples from the pillow basalts of this site 17 (location M of Figure 4) revealed magnetic components with easterly declinations and shallow negative inclinations (mean D/I = 91/-20, k = 17). The maximum blocking temperatures of these components were between 575° and 595°C and the coercivities $\lesssim$30 mT.

For the andesite porphyry of site 18, both thermal and alternating field procedures produced univectorial demagnetization trajectories such as the one shown in Figure 6c. The maximum blocking temperatures of this magnetization were approximately 585°C and the coercivities were generally $\lesssim$90 mT. The directions of this site clustered well (k = 85), and the mean has a westerly declination and a moderate negative inclination.

Sites 19A and 19B (locality K of Figure 4) were located on two sides of an isolated outcrop of a conglomerate that was striking N60°W and dipping 90°. The conglomerate was made of well-rounded basalt clasts in a hematite-stained matrix. Samples from site 19A sometimes exhibited two components of magnetization that could be more clearly resolved with thermal demagnetization. Seven of the eight samples had a component that was

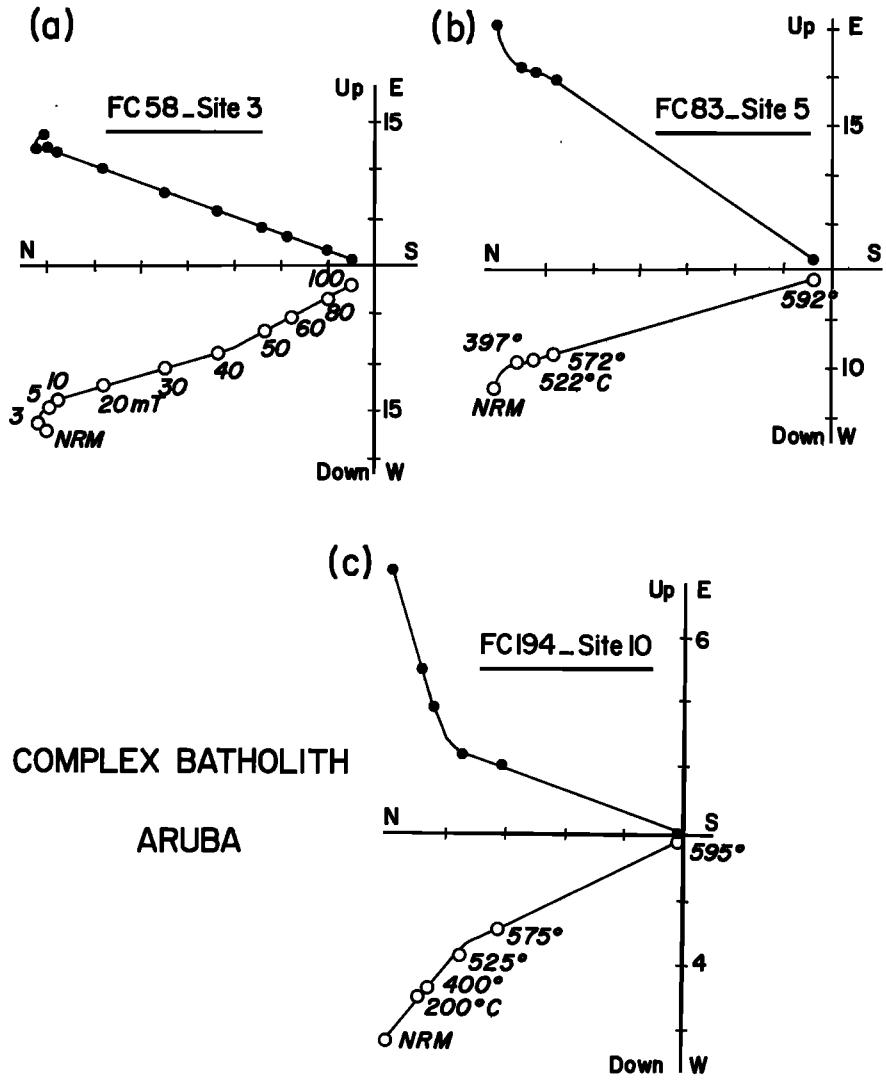


Fig. 7. Demagnetization diagrams (as in Figure 5) for samples from the batholith of Aruba. (a) An af demagnetization of a sample from locality C. (b) Thermal demagnetization of a sample from locality E. (c) Thermal demagnetization of a sample from site 10, locality G, showing two components of magnetization.

westerly with shallow positive inclinations (D/I = 280/10, k = 54). Four samples showed other components which did not cluster. All of the samples from site 19B were thermally demagnetized and yielded univectorial Zijderveld diagrams like the one in Figure 6a. The blocking temperatures for this magnetization were less than 575°-595° C. The mean directions for sites 19A and 19B are approximately antiparallel (D/I = 280/10 and 122/-20, respectively). There are two explanations for these anti parallel directions. First, the magnetizations may have been acquired during periods of opposite polarity. Second, because the sites are located on either side of an outcrop of vertically dipping strata, it might be argued that they could be located on opposite limbs of a vertical fold. This is extremely unlikely, however, as rotation about the strike (∿ 60° W) produces no significant change in the declination and inclination values and cannot bring the antiparallel directions together.

The stereographic projection of the site means in Figure 9b shows that the four sites in the diabase-schist-tuff are characterized by easterly declinations and shallow inclinations, as was also the case for the Washikemba Formation of Bonaire. Scatter in the data may be explained by the lack of knowledge about structural corrections for sites 17 and 18, although the corrections that were possible for sites 19A and 19B did not alter the data significantly.

Aruban Batholith

Most of the samples from the batholith on Aruba yielded only one characteristic component of magnetization. The site means cluster very tightly, as can be seen in Figure 9c, and the internal behavior of most of the sites was very consistent as shown by the precision parameters and the α_{95} values listed in Table 1. The samples from sites 11-13 are from a lithology with abundant xenoliths, but for a majority of samples characteristic directions could be obtained.

Typical Zijderveld diagrams for both alternating field and thermal demagnetization are shown in figures 7a and 7b, respectively. Because thermal

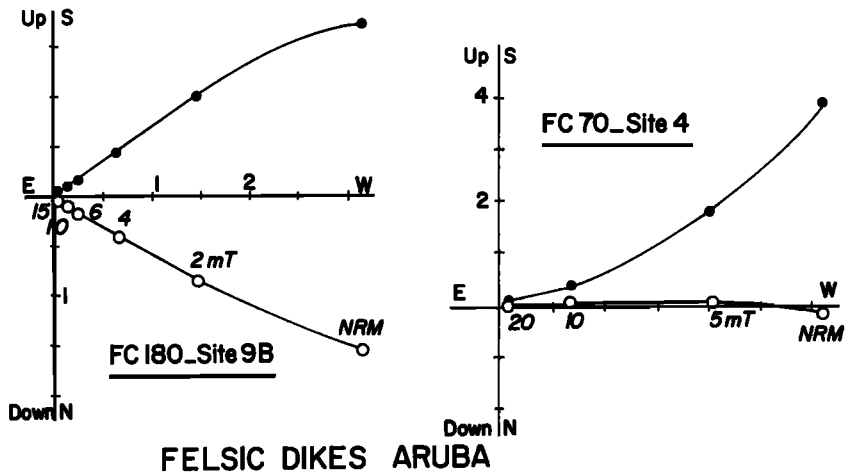


Fig. 8. Demagnetization diagrams (as in Figure 5) for af demagnetization of two felsic dikes from site 9B (locality G) and site 4 (locality D).

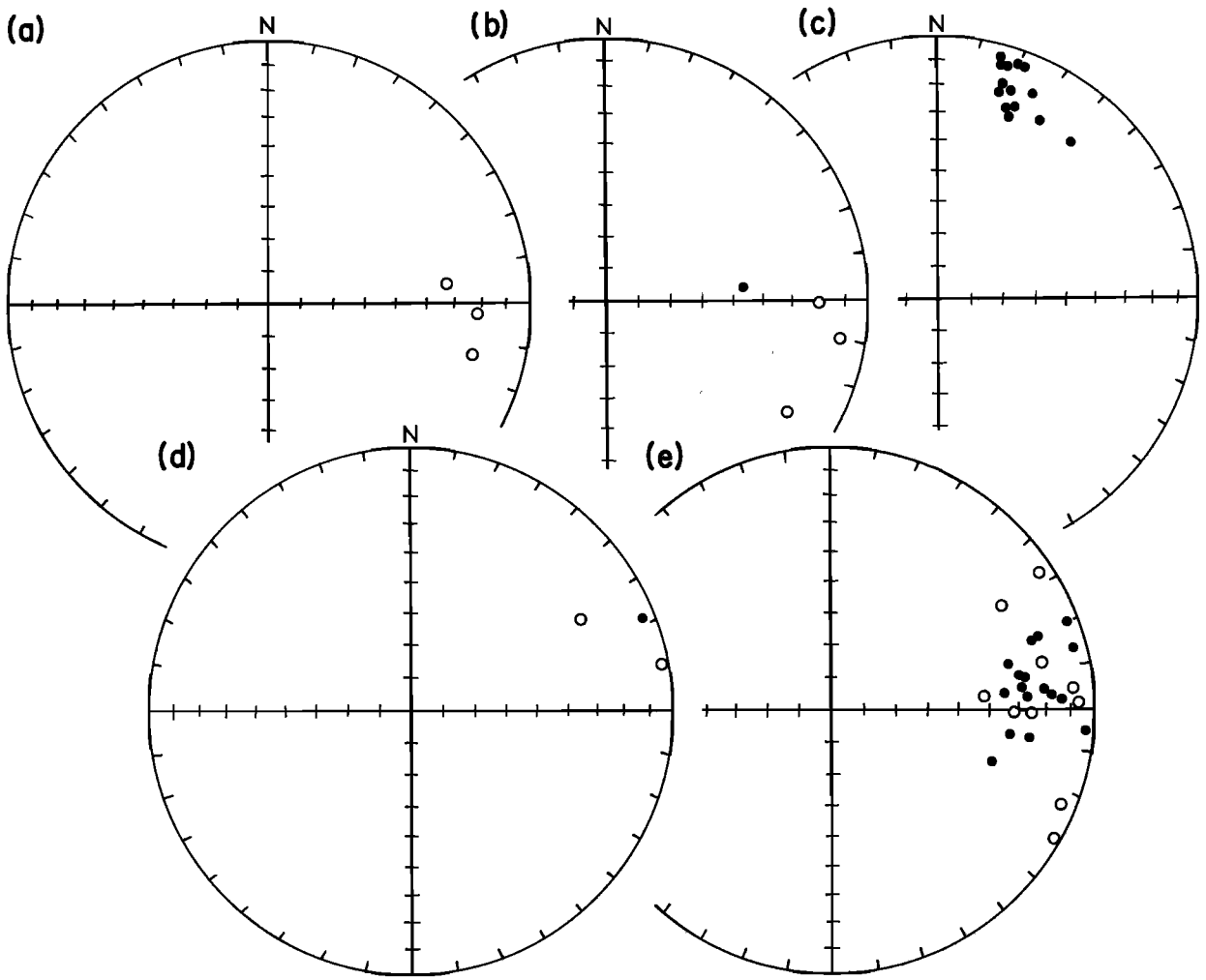


Fig. 9. Equal-area projection of the site means (all inverted to inferred normal polarity). (a) The Washikemba Formation of Bonaire. (b) The Older (Diabase-schist-tuff) Formation of Aruba. (c) The predominant magnetizations of the batholith of Aruba. (d) The felsic dikes crosscutting the Aruba batholith. (e) The first-removed directions (with lower coercivities and blocking temperatures) for 27 samples of the batholith. Solid circles are positive inclinations plotted on the lower hemisphere, and open circles represent negative (upper hemisphere) inclinations.

TABLE 1. Paleomagnetic Site Means

Map Locality	Site	Decl/Incl, deg.	N	k	α_{95}	R	Pole Position
			Aruba Batholith				
A	1	30/+23	9/10	136	4.4	8.94	61°N, 18°E
B	2	41/+23	3/10	138	10.5	2.99	50°N, 16°E
C	3	21/+24	6/7	213	4.6	5.98	70°N, 18°E
E	5	25/+15	9/9	130	4.5	8.94	65°N, 29°E
E	6	16/+ 8	7/9	39	9.8	6.85	72°N, 47°E
F	7	19/+ 6	9/9	89	5.5	8.91	69°N, 45°E
F	8A	22/+23	7/7	118	5.6	6.95	69°N, 19°E
F	8B	20/+17	9/9	168	4.0	8.95	70°N, 29°E
G	9A	17/+19	7/7	173	4.6	6.97	73°N, 27°E
G	10	22/+27	7/7	125	5.4	6.95	69°N, 13°E
H	11	201/- 5	6/7	7	27.6	5.27	67°N, 44°E
H	12A	16/+ 8	5/7	39	12.3	4.90	72°N, 47°E
H	12B	17/+17	6/7	20	15.5	5.74	73°N, 31°E
H	13A	15/+ 5	5/10	34	13.3	4.88	72°N, 53°E
			Aruba Felsic Dikes				
D	4	249/- 5	9/10	42	8.0	8.81	21°N, 22°E
F	8B	80/- 2	7/8	13	17.4	6.54	10°N, 23°E
G	9B	242/+28	8/8	56	7.5	7.87	23°N, 42°E
			Aruba—Older Formation				
M	17	91/-20	15/20	17	9.7	14.15	3°S, 30°E
L	18	265/-46	12/12	85	4.7	11.87	10°N, 354°E
K	19A	280/+10	7/8	54	8.3	6.88	11°S, 22°E
K	19B	122/-20	5/6	112	7.3	4.96	33°S, 24°E
			Bonaire-Washikemba Formation				
B	20	104/-21	11/11	90	4.9	10.89	16°S, 30°E
C*	21	26/-30	11/11	4	25.1	8.66	-- --
D	22	273/+21	9/9	58	6.8	8.86	5°S, 30°E
A	23	84/-32	9/13	386	2.6	8.98	2°N, 38°E

Decl/Incl are declination and inclination after demagnetization; N is the ratio of samples used in the calculations to samples analyzed; k, α_{95}, and R are parameters of the statistical calculation [McElhinny, 1973].
* The data for this site are disregarded because of lack of within-site consistency.
Map locations refer to Figures 3 and 4.

demagnetization was generally more successful, most of the samples were treated thermally. Alternating field demagnetization was done on at least one sample per site, but often this technique did not sufficiently break down the magnetization. The blocking temperatures were less than 575°-595°C.

All of the samples from site 10 and 20 samples from various other sites yielded two characteristic components. A typical Zijderveld diagram from the data from site 10 is shown in Figure 7c. The first component was removed by 400°C, and its mean for this site is easterly with a shallow positive inclination (mean D/I = 84/24). The second component had blocking temperatures up to 575°-595°C and matches the directions from the other sites in the batholith (mean D/I = 22/27). Both of these directions clustered very well. The softer directions of site 10 are included in the plot (Figure 9e) of all first-removed easterly components observed in 27 samples of the batholith. The importance of the first-removed direction will be discussed below.

Two felsic dikes, sites 4 and 9B, which crosscut sites 5 and 9A, respectively, yielded univectorial Zijderveld diagrams upon thermal as well as alternating field demagnetization. Examples are shown in Figure 8. Both of the sites clustered well (k = 42 and 56), and the site mean declinations are westerly with shallow inclinations. A third dike (site 8B), perhaps the continuation of the dike of site 9B, yielded a nearly opposite direction to that observed at sites 4 and 9B. The samples of this felsic dike appeared to be more altered, which may account for their much lower intensities of magnetization. The dike directions are shown in Figure 9d.

The projection of the characteristic site means

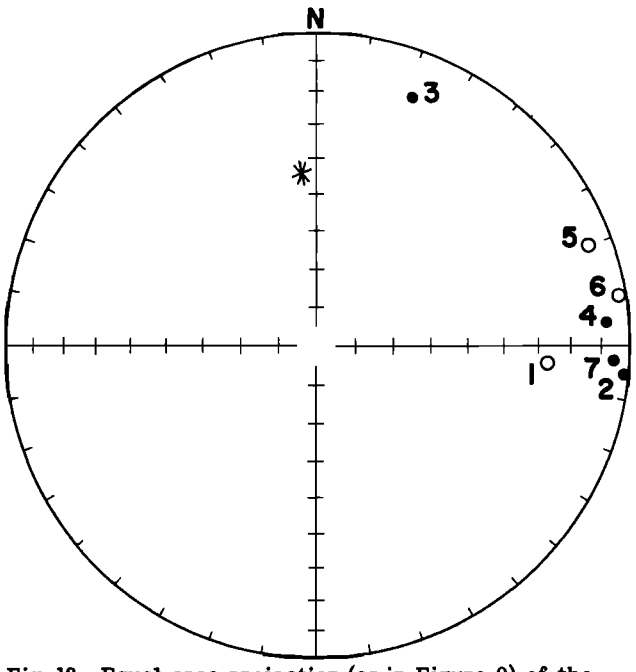


Fig. 10. Equal-area projection (as in Figure 9) of the mean paleomagnetic directions for the Bonaire block (as listed in Table 2). Also plotted is the present-day geomagnetic field direction in Aruba (asterisk).

of Figure 9c and that of the softer directions (Figure 9e) shows that the Aruban batholith is characterized by two directions. Most of the samples yield univectorial directions that are northeasterly with shallow positive (down) inclinations. However, the data from the three felsic dikes that crosscut the batholith (Figure 9d) and the first-removed directions seen in 27 of the 115 samples of the batholith (Figure 9e) are easterly with very shallow inclinations.

Discussion of the Results

The means of the site means for the Washikemba Formation, the diabase-schist-tuff formation, the prevalent directions of the Aruban batholith, the softer directions of the Aruban batholith, and for the felsic dikes that cut this batholith along with the paleomagnetic results of MacDonald and Opdyke [1972] for Guajira, Colombia, and the paleomagnetic results of Skerlec and Hargraves [1980] for the Caribbean Mountains of northern Venezuela, are plotted in an equal area projection in Figure 10 and are tabulated in Table 2. Two distinct paleomagnetic directions can be seen in our data from Aruba and Bonaire. One, with easterly declinations and shallow inclinations, characterizes the Washikemba Formation of Bonaire, three felsic dikes that crosscut the Aruban batholith and the diabase-schist-tuff formation on Aruba. This direction is also seen in samples of the Aruba batholith, as low blocking-temperature components. These results are similar to the data from Guajira, Colombia, and from the Caribbean Mountains. The second direction, approximately N20° E, with a shallow positive inclination, characterizes the bulk of the samples from the batholith on Aruba. The correlation of the easterly directions between sites that are widely separated indicates that plausible structural corrections would not alter this bimodal grouping significantly.

The ages of the magnetizations which yielded the easterly directions of Figure 10 are all inferred to be Late Cretaceous/early Paleocene or older. The La Teta lavas of Guajira (Colombia) are earliest Cretaceous or older. The results for northern Venezuela have a wide age range spanning the Cretaceous. The magnetizations of the Washikemba Formation and the diabase-schist-tuff formation might be primary, but the lack of a fold test and the low-grade (greenschist facies) metamorphism of these rocks could argue for a secondary origin of the magnetizations, related to the Late Cretaceous/early Paleocene thermal event and the resetting of the K-Ar ages mentioned earlier. The paleomagnetic and geologic evidence does not support a magnetization age younger than that of this event.

We interpret these data as an indication of a tectonic rotation of the Bonaire Block. This is also illustrated in Figure 11, which is an orthographic projection of the paleomagnetic poles calculated from our data and the published poles for cratonic South America [Creer, 1962, 1970; Valencio, 1972; Opdyke and MacDonald, 1973; Pacca and Hiodo, 1976; Vilas, 1976; Valencio et al., 1977; Mendia, 1978]. In general, the older rocks of the Bonaire Block carry easterly directions, and younger rocks (the Aruba batholith) carry northeasterly directions, thus suggesting a clockwise rotation of approximately 90° since the Late Cretaceous/early Paleocene. When Silver et al. [1975] introduced the idea of the Bonaire Block, they suggested that a clockwise rotation would best explain the nonuniform deformation of the Curacao Ridge and the lack of any similar deformation east of the Los Roques Canyon. If we had only the easterly directions, the sense of the rotation would not be determinable from the paleomagnetic data. Because there are a number of magnetic reversals in the Late Cretaceous and early Tertiary [Lowrie and Alvarez, 1977; Roggenthen and Napoleone, 1977] and because we do not have precise ages for our samples which have low (near-equatorial) inclinations, we cannot determine whether the magnetizations they carry were acquired during a normal or reversed period. If the magnetizations were acquired during a reversed period, the easterly directions would imply a counterclockwise rotation. However, the northeasterly direction yielded by the Aruban batholith, which we interpret to be a Late Cretaceous/early Paleocene direction, is intermediate between an easterly (rotated) direction and an unrotated northerly direction. Because of this, we argue that the paleomagnetic data suggest a clockwise rotation.

One big problem with this interpretation is that three felsic dikes (sites 4, 8B, and 9B), which cut the batholith and must therefore be younger, apparently carry what we consider to be the earlier Cretaceous direction. Two explanations are possible. If all the magnetizations are primary, then our data would support a three-phase rotational history for the Bonaire Block. More specifically, they would suggest a clockwise rotation followed by a counterclockwise rotation which in turn was followed by a clockwise rotation. An alternative explanation is that the magnetic directions of the Aruban batholith are of 'secondary' origin acquired during chemical alternation or a low-grade metamorphic event sometime after the initial cooling of the rocks and

TABLE 2. Summary of Paleomagnetic Data for the Bonaire Block

Number	Unit Studied	N	Decl/Incl, deg.	k	α_{95}	R	Pole Position	Reference
1	Bonaire-Washikemba Fm.	29 samples	94.3/-27.1*	43.6	4.1	28.36	7.1°S, 33.2°E	this study
		3 sites	93.6/-27.7*	54.1	16.9	2.96	6.5°S, 33.7°E	this study
2	Aruba-Older Formation	39 samples	95.4/+ 1.4	5.7	10.5	32.35	5.1°S, 18.1°E	this study
		4 sites	100.4/- 2.3	5.5	43.0	3.45	10.4°S, 18.9°E	this study
3	Aruba-batholith (predominant direction)	14 sites	21.4/+15.8	62.6	5.1	13.79	68.5°N, 29.7°E	this study
4	Aruba-batholith ('soft' direction)	27 samples	86.1/+ 8.0	9.7	9.4	24.32	4.8°N, 16.9°E	this study
5	Aruba-felsic dikes	24 samples	70.2/- 8.2	14.0	8.2	22.36	18.3°N, 28.7°E	this study
		3 sites	70.7/- 8.3	17.4	30.5	2.89	17.9°N, 28.6°E	this study
6	Colombia-lavas and diorite of the Guajira peninsula	9 sites	80.5/- 0.3*	9.1	18.0	-	9.2°N, 22.2°E	MacDonald and Opdyke [1972]
7	Venezuela-Villa de Cura, Cerro Pelon and Tinaco belts	15 sites	93.2/+ 5.6*	5.1	18.8	12.25	2.5°S, 16.6°E	Skerlec and Hargraves [1980]

Numbers correspond to those of Figure 10; N is the number of statistical entires in the analysis; k, α_{95}, and R are the statistical parameters [McElhinny, 1973].

*Directions have been recalculated from the pole positions for the coordinates of Aruba (12.45° N, 70.0° W).

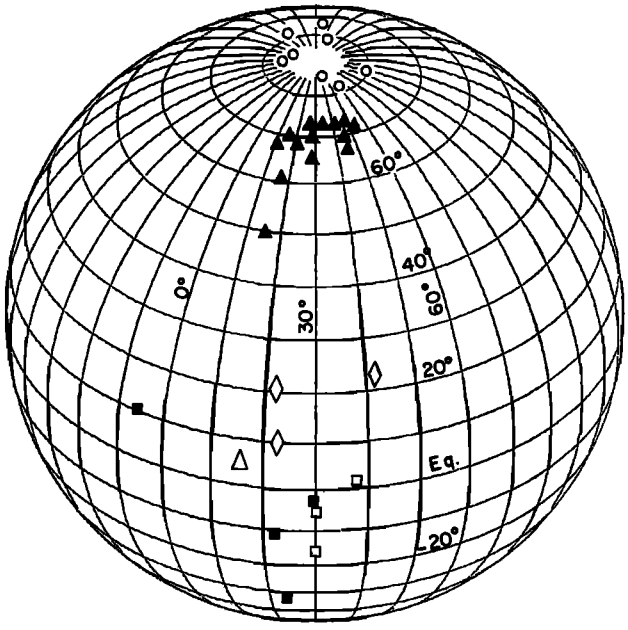


Fig. 11. Pole positions derived from the site-mean directions from the Washikemba Formation (open squares), the Older (Diabase-schist-tuff) Formation of Aruba (solid squares), the felsic dikes (open diamonds), the two directions of the Aruba batholith (solid and open triangles), and poles for the South American craton (open circles).

after the emplacement of the three felsic dikes. The idea of a chemical/mineralogical alteration is supported by the low-blocking temperature magnetic components seen in the data from the batholith; these could be remnants of the original magnetization.

The critical question then is whether the magnetization carried by the batholith is primary or secondary. To look more closely at the problem, we have examined the mineralogy and the magnetic properties of the samples from the batholith in more detail. Thermomagnetic curves with Curie temperatures of about 570°C on several of our samples indicate that magnetite is the principal carrier of the remanence. Hence, the discussion involves primarily the magnetite-ulvospinel solid solution series. In this series, there is complete solid solution at temperatures greater than 600°C. At lower temperatures the phases tend to exsolve. If oxidation occurs, some or all of the original spinel phases tend to be oxidized toward the titano-maghemite or ilmenite-hematite series.

Reflected-light microscopy indicates an abundance of apparently pure and primary magnetite, some smaller grains of ilmenite and numerous grains with exsolution features of magnetite-ilmenite in the rocks from the batholith. Initially, we explored the possibility that the original mineralogy was a titanomagnetite and hypothesized that this phase could have carried a primary Cretaceous direction of magnetization. If, sometime after most of the tectonic rotation had occurred and after the intrusion of the felsic dikes, exsolution and oxidation occurred, producing the magnetite and ilmenite that can be seen in the rocks, the original magnetization could have been partially to completely replaced by a younger magnetization. The rationale for this hypothesis was provided by, for instance, the behavior of the samples from site 10. As the Zijderveld diagram in Figure 7c illustrates, these samples have two components of magnetization, an easterly direction with blocking temperatures less than 400°C and a northeasterly component with blocking temperatures less than 575°-595°C but higher than 525°C. The blocking temperatures might suggest that the easterly direction is carried perhaps by a titanomagnetite and the northeasterly direction by a relatively pure magnetite. The fact that only these samples from site 10 and 20 samples from other sites revealed both components can be explained by the hypothesis that exsolution in these rocks was not as complete.

However, extensive thermomagnetic analyses and X-ray studies do not support this hypothesis because no evidence was found for titanomagnetites (with Curie temperatures in the range of 300°-500°C).

Since the northeasterly direction is so pervasive in the Aruban batholith, we favor a metamorphic-hydrothermal-rock magnetic explanation rather than invoking atypical geomagnetic behavior or explanations involving polarity transitions, but we cannot prove our contentions. Helmers and Beets [1977] reported that hydrothermal alteration of the batholith has been important. In some thin sections we have seen unambiguous secondary magnetite of very small grain size formed by alteration of amphiboles, but we are left with no explanation for the question why secondary magnetites would be the prevailing carriers of the remanence when clearly primary magnetites are quite abundant.

Despite the fact that we do not have a reasonable mineralogical explanation for the idea that the batholith carries a magnetization younger than that of the dikes which intruded the batholith, we prefer the interpretation of a simpler, single rotation of the Bonaire Block. It is interesting to note, however, that Gose and Schwartz [1977] advocated a complex to and fro Late Cretaceous rotation of the Honduras microplate, which is similar to our less preferred hypothesis of multistage rotations. The acquisition of more well-dated results from the Venezuelan continental borderland might reveal a more complex rotational history for the Bonaire Block in the future.

Skerlec and Hargraves [1980] discuss two possible models for the rotation of the Bonaire Block. One describes the rotation of a single coherent microplate about one rotation axis, the other involves numerous in situ rotations of much smaller crustal blocks. They suggest that the regional geology of the Caribbean Mountains and their paleomagnetic data are more reasonably explained by a single microplate rotation. The correlation of our data with theirs and those of MacDonald and Opdyke [1972], which all indicate a 90° rotation with respect to cratonic South America regardless of the variations in age of the magnetizations, is additional evidence for a single rotation of a coherent crustal block. It is less likely that small microplates rotating independently would produce paleomagnetic data that correlate so well.

Paleolatitudes for Aruba and Bonaire were calculated from the inclinations of the two characteristic groups of directions in our data and then compared with paleolatitudes calculated for Aruba and Bonaire using the cratonic South American poles (Figure 11) as predictors. The inclination of the

mean of the formation means with easterly directions, as shown in Figure 10, is -6.5° (N = 4, α_{95} =22.2°) for Aruba and Bonaire only, based on the entries 1, 2, 4, and 5 of Table 2, whereas it is -3.4° (N = 6, α_{95} = 13.5°) for the entire Bonaire Block (mean of entries 1, 2, and 4-7 of Table 2). This yields paleolatitude values of 3°S ± 11° and 2°S ± 7°, respectively, whereas the mean of the Aruba batholith gives a paleolatitude of 8°N ± 2.5°. Furthermore, the mean paleolatitude calculated from published Cretaceous South American poles for Aruba is 12.3°N , which is similar to today's value. Although perhaps not statistically significant, we interpret the changes in paleolatitude with respect to cratonic South America as a relative northward drift of the Bonaire Block since the Cretaceous, in addition to paleomagnetically undetectable west-east movements. Although the relative northward drift is small, it is comparable to an earlier prediction [MacDonald and Opdyke, 1972] for the Guajira Peninsula in Colombia of 5°-10° northward displacement relative to cratonic South America.

In conclusion, we interpret the results of a paleomagnetic investigation of Late Cretaceous and early Tertiary rocks from Aruba and Bonaire as evidence for a 90° clockwise rotation of the Bonaire Block in addition to a small northward drift as well as possibly west-east movements. If the predominant magnetizations in the Aruban batholith are indeed secondary, then our data suggest that the rotation remains unbracketed in time but must have occurred sometime after the initial cooling of the batholith more than 70 my ago.

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